Frontend Development

Study Notes – Definitions and Answers

1. What is Frontend Development?

Definition: Frontend development is the process of creating the part of a website or web application that users can see and interact with directly. It focuses on the layout, design, buttons, menus, forms, images, text, and overall user experience.

Explanation: When a user opens a website, the visible page is the frontend. A frontend developer writes code to make the page attractive, easy to use, responsive, and interactive. For example, the navigation bar, login form, product cards, and buttons on an online shopping website are part of its frontend.

Main objectives of frontend development:

- Create a clear and attractive user interface (UI).
- Make websites easy to use and navigate.
- Ensure pages work on mobiles, tablets, and computers.
- Add interaction, such as menus, form validation, and animations.
- Make pages load efficiently and work across common browsers.

2. Difference Between Frontend and Backend

Frontend and backend are two important parts of web development. The frontend is the user-facing part, while the backend works behind the scenes to process data and provide services.

Frontend	Backend
The part of a website that users see and interact with.	The part that works behind the scenes and users do not normally see.
Focuses on design, layout, navigation, and interaction.	Focuses on server logic, databases, authentication, and data processing.
Common technologies: HTML, CSS, JavaScript, React, and Angular.	Common technologies: Node.js, Python, Java, PHP, and databases such as MySQL.
Runs mainly in the user's web browser.	Runs mainly on a web server or cloud environment.
Example: displaying a login form and checking that required fields are filled.	Example: checking login details against stored account data.

Example: In an online shopping website, the product page, images, filters, and Add to Cart button are frontend features. Retrieving product details, calculating the order total, processing payment securely, and saving the order are backend tasks.

3. Overview of Web Technologies

Definition: Web technologies are the languages, tools, frameworks, and systems used to create, run, and maintain websites and web applications.

Important web technologies include:

HTML (HyperText Markup Language): Defines the structure and content of a webpage, such as headings, paragraphs, images, links, and forms.

CSS (Cascading Style Sheets): Controls the appearance of a webpage, including colours, fonts, spacing, borders, and page layout.

JavaScript: Adds behaviour and interactivity, such as dropdown menus, sliders, form checks, and content that updates without reloading the whole page.

Frontend frameworks and libraries: Tools such as React, Angular, and Vue help developers build and manage complex user interfaces.

Backend technologies: Technologies such as Node.js, Java, Python, PHP, and .NET handle server-side operations and application logic.

Databases: Systems such as MySQL, PostgreSQL, and MongoDB store and organize information, such as user accounts and product details.

Web servers and hosting: Software and services that make websites available to users over the internet.

HTTP and HTTPS: Communication protocols used by browsers and servers to exchange information. HTTPS encrypts the connection.

How they work together: HTML provides the page structure, CSS styles it, and JavaScript makes it interactive. The browser communicates with backend services through web protocols, while the backend can read from or write to a database.

4. Roles and Responsibilities of a Frontend Developer

Definition: A frontend developer is a professional who builds and maintains the user-facing part of websites and web applications. The developer turns designs and requirements into working webpages.

- Build webpage structure using HTML.
- Style pages using CSS and responsive layout techniques.
- Use JavaScript to add interactive features.
- Convert UI/UX designs and prototypes into functional webpages.
- Make websites responsive so they work on different screen sizes.
- Test pages in different browsers and fix layout or functionality issues.
- Improve performance so pages load smoothly and efficiently.
- Make websites accessible and usable for people with different needs.
- Connect the frontend to backend services and APIs to display or submit data.
- Maintain existing code, fix bugs, and work with designers, backend developers, and testers.

Important skills: HTML, CSS, JavaScript, responsive design, browser developer tools, basic accessibility, debugging, version control (such as Git), and communication/teamwork.

Quick Revision

- **Frontend:** The visible and interactive part of a website.
- **Backend:** The server-side part that processes information and manages data.
- **HTML:** Structure of a webpage.
- **CSS:** Style and layout of a webpage.
- **JavaScript:** Behaviour and interactivity of a webpage.
- **Frontend developer:** Builds, tests, improves, and maintains the user-facing interface.